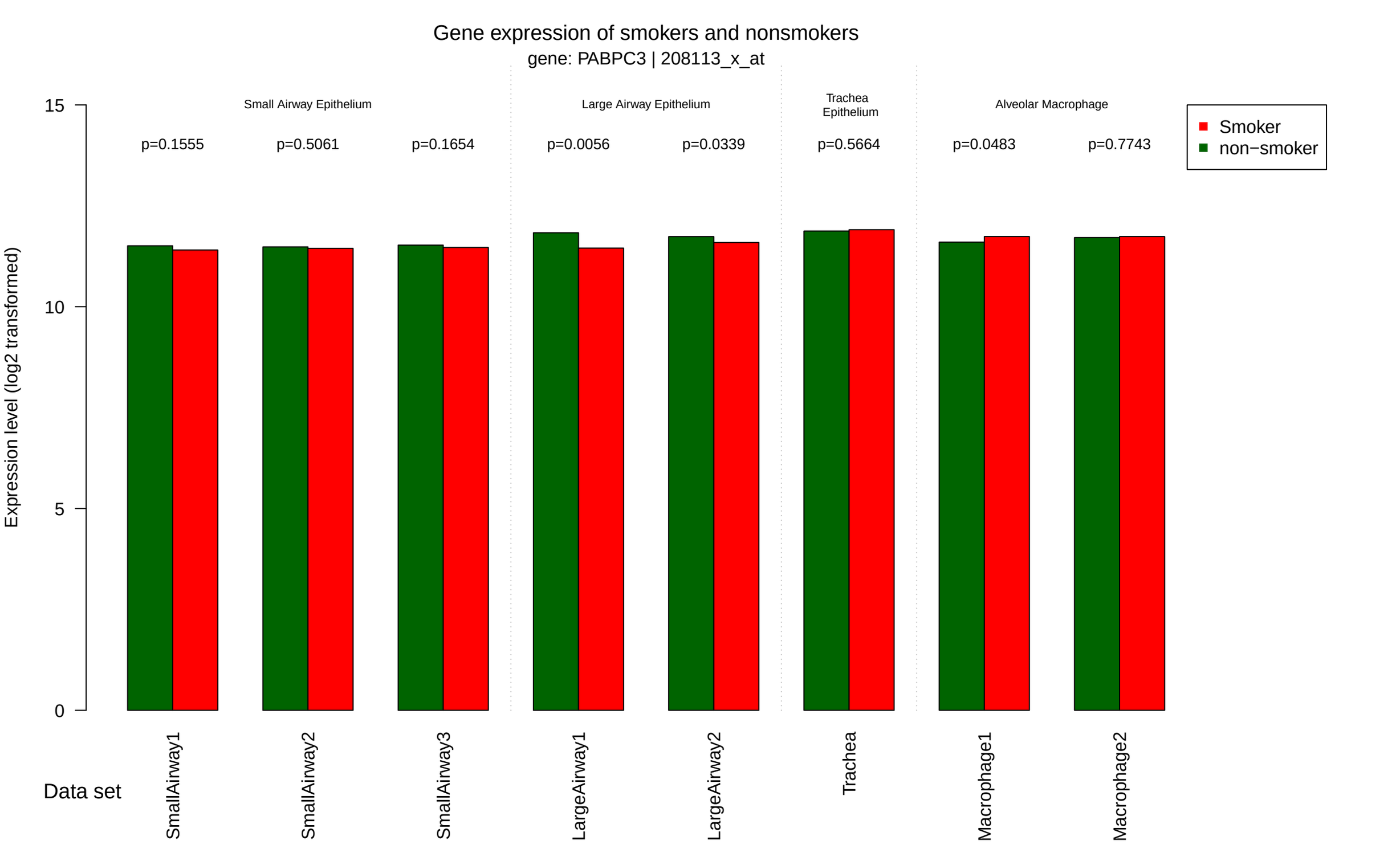
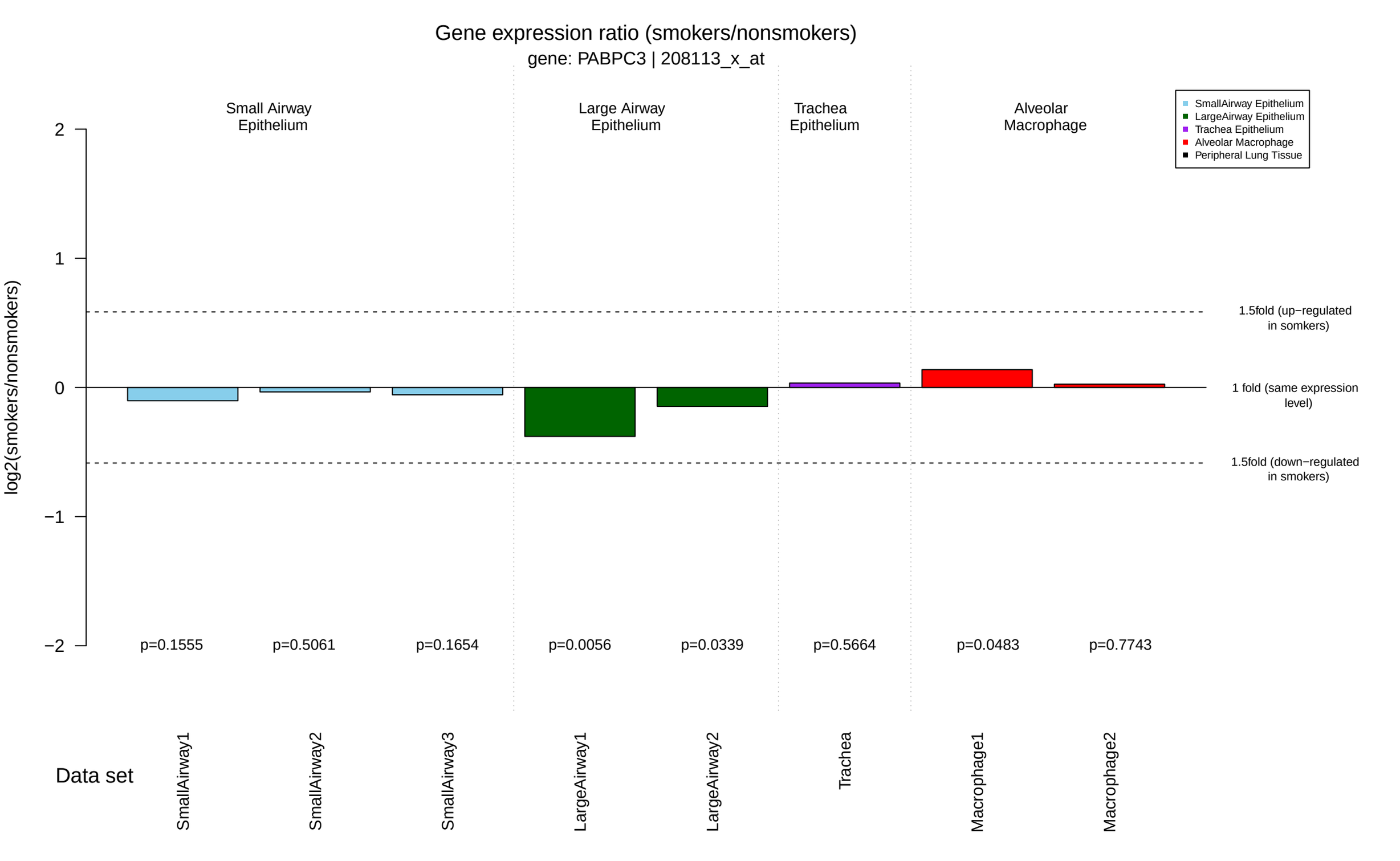
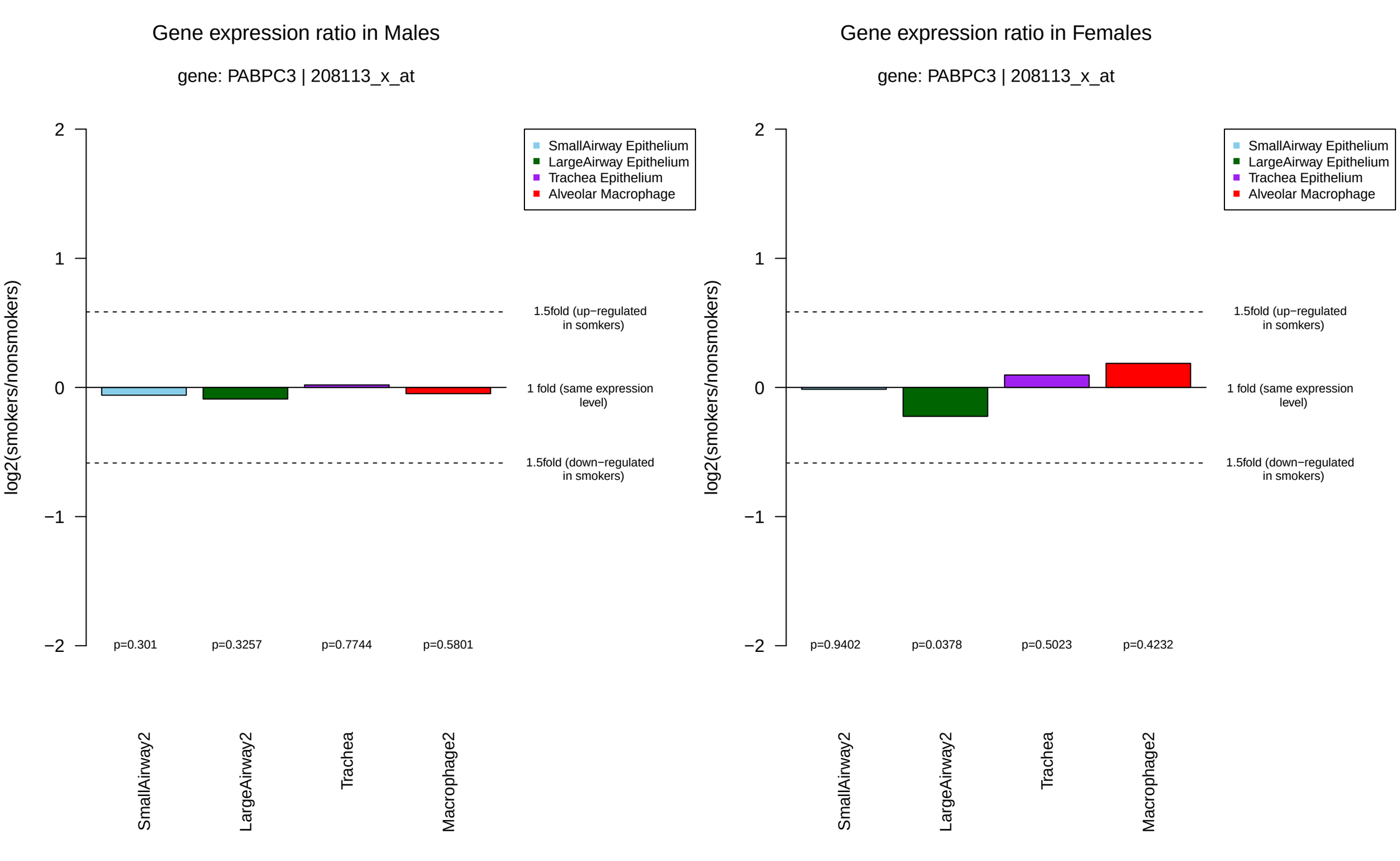
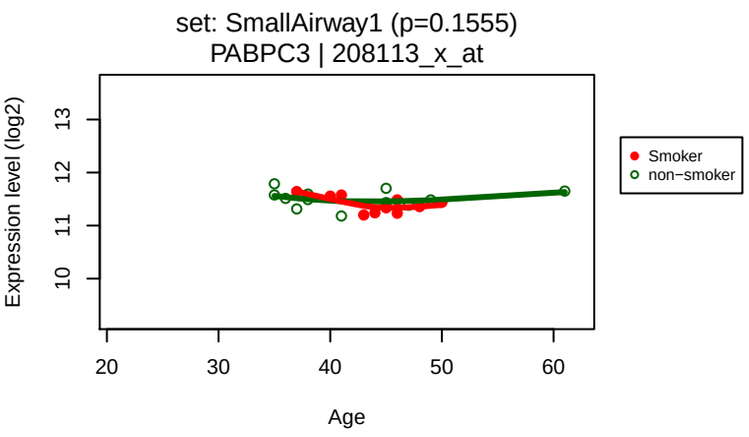
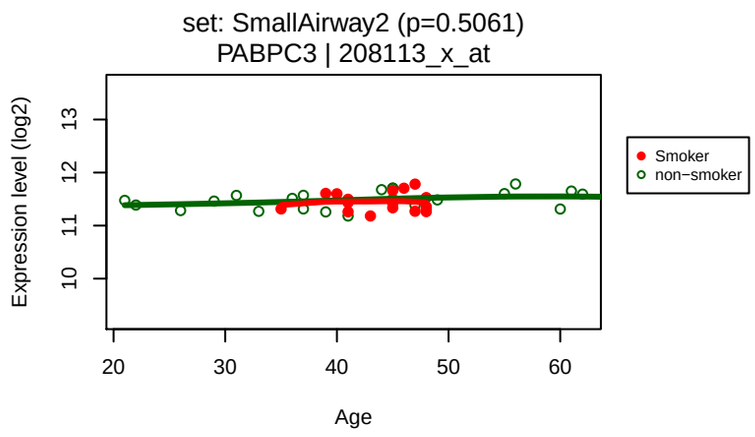
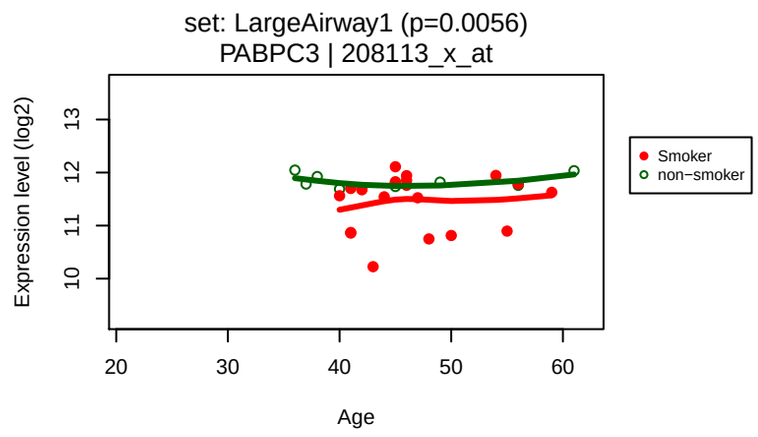

Gene expression of smokers and nonsmokers
gene: PABPC3 | 208113_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1555
p=0.5061
p=0.1654
p=0.0056
p=0.0339
p=0.5664
p=0.0483
p=0.7743
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

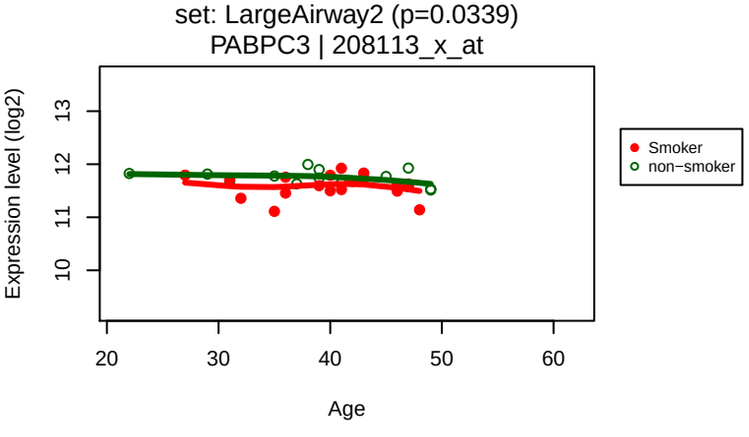

Gene expression ratio (smokers/nonsmokers)
gene: PABPC3 | 208113_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1555
p=0.5061
p=0.1654
p=0.0056
p=0.0339
p=0.5664
p=0.0483
p=0.7743
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

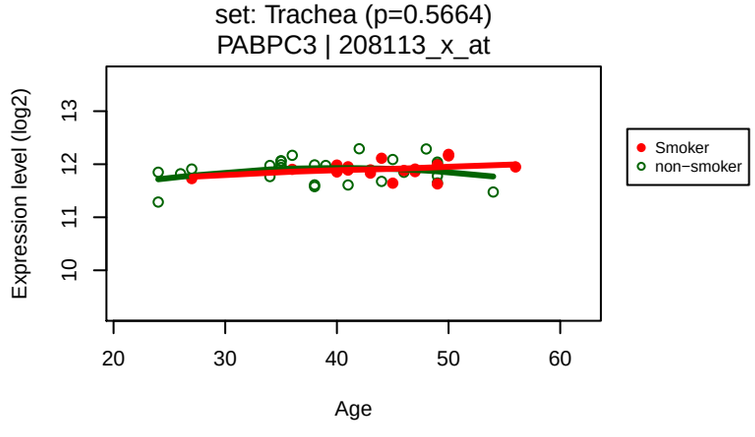

Gene expression ratio in Males
gene: PABPC3 | 208113_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.301
p=0.3257
p=0.7744
p=0.5801
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PABPC3 | 208113_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9402
p=0.0378
p=0.5023
p=0.4232
SmallAirway2
LargeAirway2
Trachea
Macrophage2

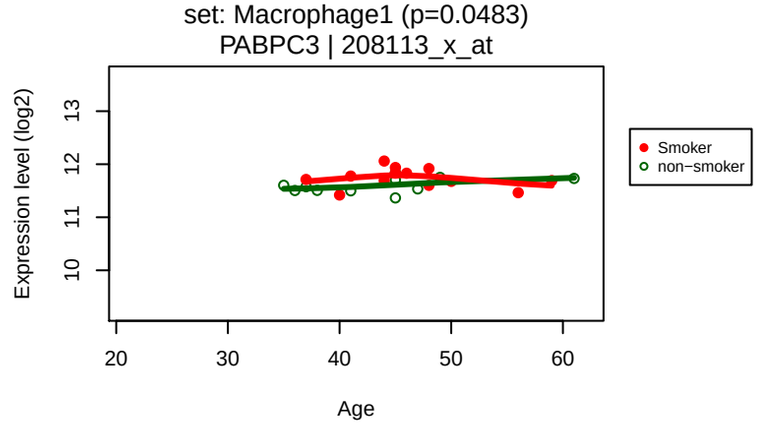

set: SmallAirway1 (p=0.1555)
PABPC3 | 208113_x_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker

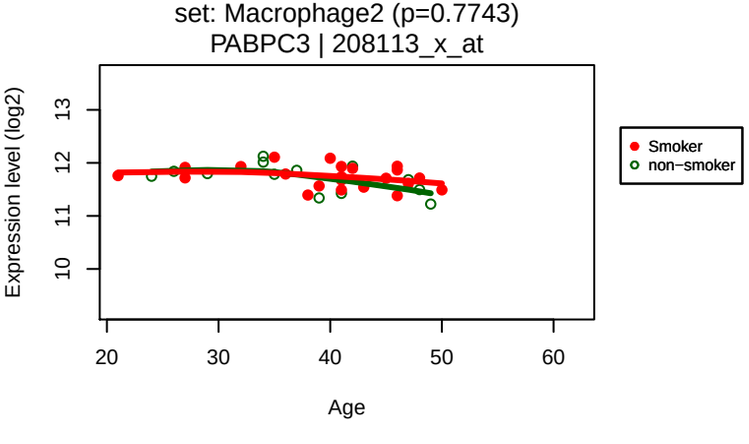

set: SmallAirway2 (p=0.5061)
PABPC3 | 208113_x_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker

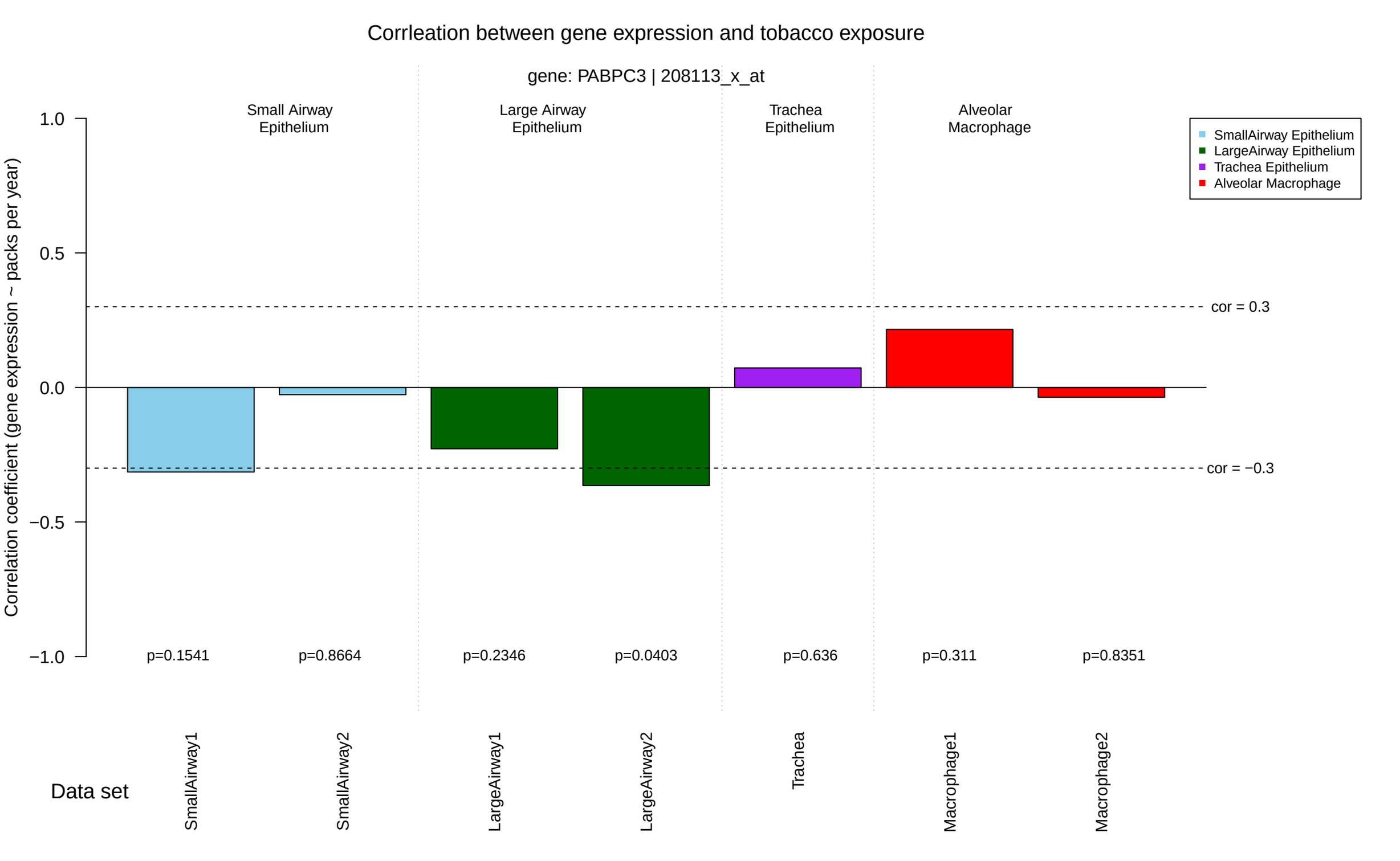

set: LargeAirway1 (p=0.0056)
PABPC3 | 208113_x_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0339)
PABPC3 | 208113_x_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.5664)
PABPC3 | 208113_x_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0483)
PABPC3 | 208113_x_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7743)
PABPC3 | 208113_x_at
Expression level (log2)
10
11
12
13
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PABPC3 | 208113_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1541
p=0.8664
p=0.2346
p=0.0403
p=0.636
p=0.311
p=0.8351
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2